

- If no such row exists, since there are no free variables, by Theorem 2, the system of eqns has a unique solution.

∴ For each b in $\mathbb{R}^m$ the equation $Ax=b$ has at most one solution $\square$

1.8 Introduction to Linear Transformation

- The difference between a matrix equation $Ax=b$ and the associated vector eqn. $x_1a_1 + \dots + x_na_n = b$ is merely a matter of notation.
- However a matrix equation $Ax=b$ can arise in linear algebra (and in applications such as computer graphics and signal processing) in a way that is not directly connected with linear combinations of vectors.
- This happens when we think of the matrix A as an object that "acts" on a vector x by multiplication to produce a new vector called Ax .

- For instance, the equations

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 A x b A u 0

say that multiplication by A transforms x into b and transforms u into the zero vector.